

Business Process Automation

with Power Automate and Copilot Studio Agents

WSQ Course Code: TGS-2022017524 · 2-Day Hands-On Training · Modules 1–4

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W
Version 5.0 (25 Jul 2026) · www.tertiarycourses.com.sg · <https://lms-tms.tertiaryinfotech.com/>

Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

CERTIFICATION

Microsoft Power Platform, AI and data analytics — 20+ years of training experience.

DELIVERS

WSQ courses on AI, business automation, data science and software engineering.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

Let's Know Each Other

- Your name and organisation / role.
- Your experience with Power Automate, Copilot Studio or other automation tools (if any).
- One repetitive business process you would love to automate after this course.

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

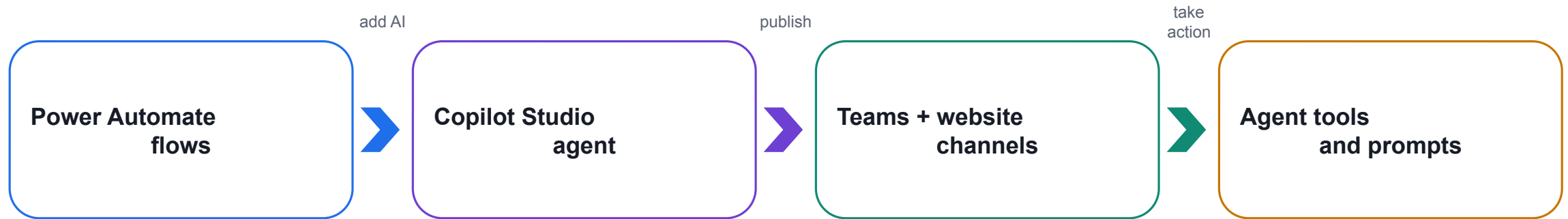
5

Be punctual; return from breaks on time.

6

75% attendance is required.

What You'll Learn



Day 1 — automate

Build reliable trigger → action flows without an AI agent.

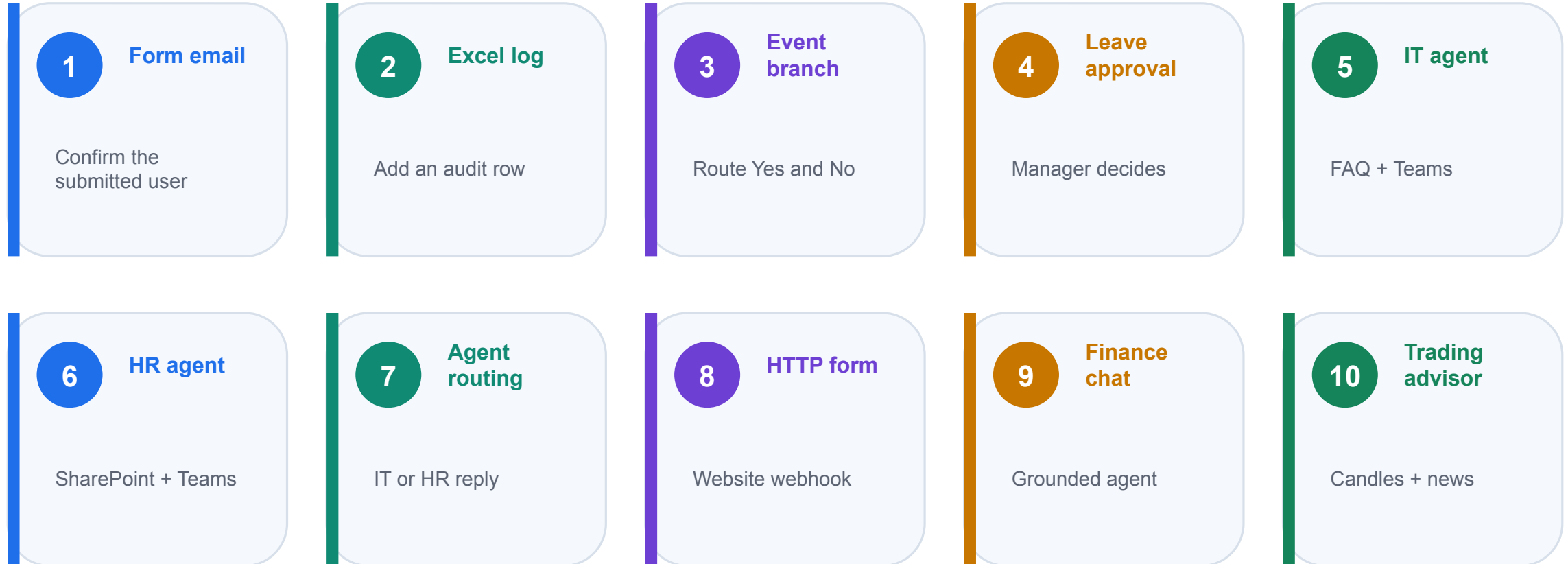
Day 2 — converse

Create a grounded agent that understands and confirms user intent.

Integrate — act

Let the agent call deterministic or prompt-based Power Automate tools.

Your 2-Day Journey



Day 1 builds Forms, flows and support agents. Day 2 adds webhooks and the Finance Advisor capstone.

How This Course Works

Concept, then demo, then hands-on — for every topic

Built entirely in Microsoft 365 and the Power Platform

No coding required — low-code, drag-and-drop tools

Save your flows and agents to reuse in your own role

- Daily recap and open Q&A to lock in what you learned

Lesson Plan — 2 Days, 9:00am–6:00pm

D1 Day 1 — Build and route

9:00–10:40 Concepts + flow types
10:55–3:15 Labs 1–4: Forms, Excel, event branch, approval
3:30–5:50 Agent concepts + Labs 5–7
5:50–6:00 Evidence check and recap

D2 Day 2 — Connect and advise

9:00–10:05 HTTP requests and webhooks
10:05–12:15 Lab 8 + Finance Agent foundation
1:15–4:00 Labs 9–10: web chat and trading advisor
4:00–6:00 Written + practical assessment

Briefing for Assessment

- Place phones and other materials under the table or on the floor.
- No photos or recording of assessment scripts.
- No discussion during the assessment.
- Use a black/blue pen for hard-copy assessments.
- No liquid paper / correction tape.
- Scripts are collected when time is up.

Assessment & Funding

1

Written

1 hour · open book
Modules 1–4 and the lab concepts

2

Practical

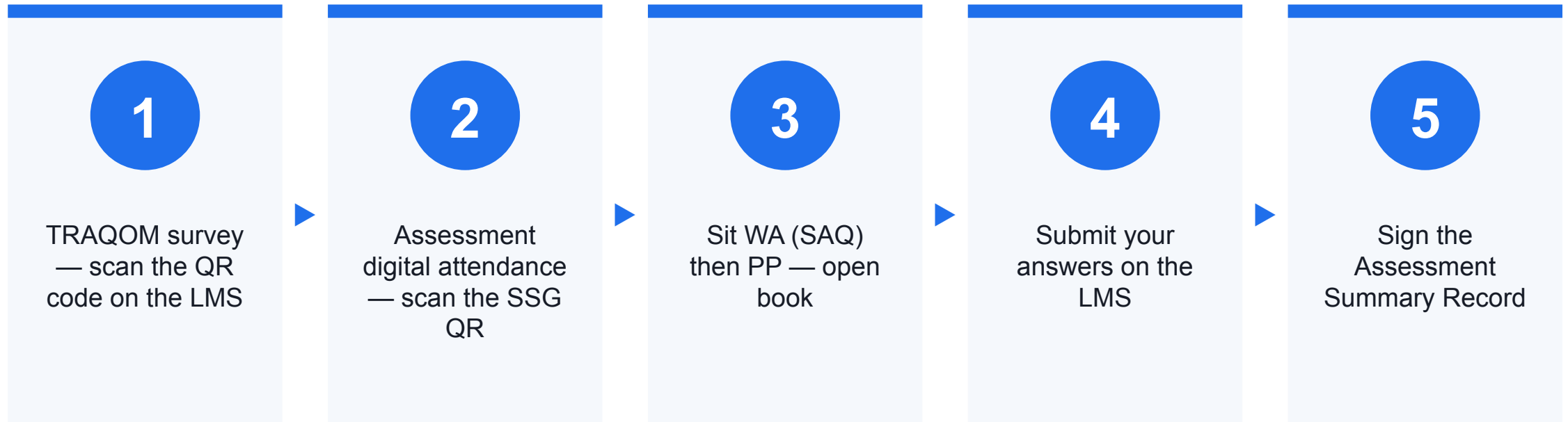
1 hour · open book
Build and verify a working flow and agent

3

Evidence

Submit on the LMS
Competent result + attendance requirement

Assessment Flow



Access the Hands-On Labs

1

10 labs

Lab 0 setup plus Labs 1–10 and four concept modules.

2

Ready assets

Excel tables, searchable PDFs, HTML pages and JSON schemas.

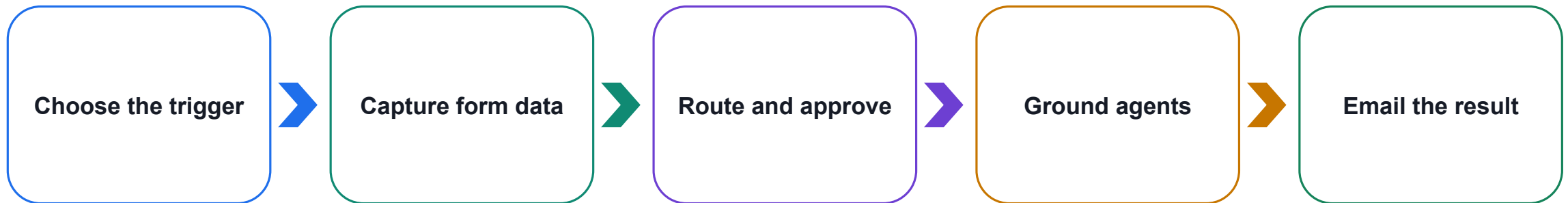
3

Use the guide

Open each index.md or the compiled Learner Guide and work in order.

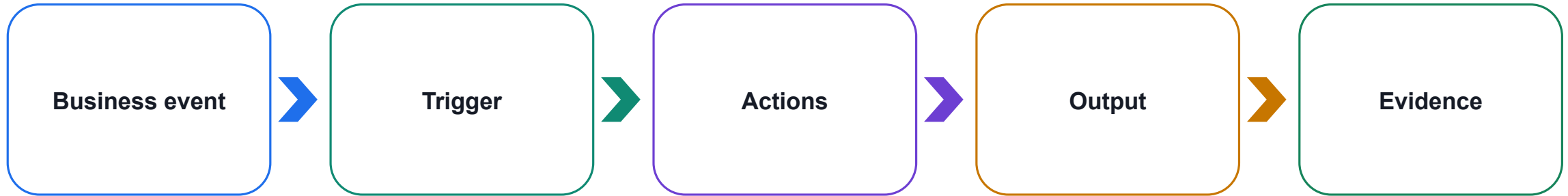
Day 1 — Forms, Flows and Business Agents

DAY 1



Build one connected automation story, then add specialised agents.

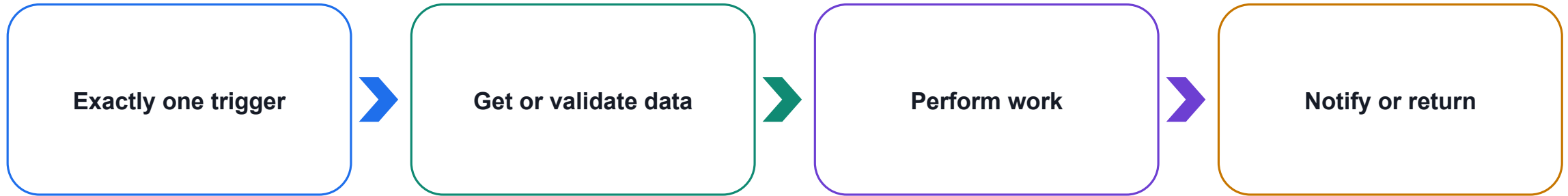
A Business Workflow Has a Clear Start and Result



Teaching point

Name the start event, required work, expected result and proof before opening the designer.

Trigger → Actions → Output



Teaching point

Every lab uses the same skeleton; only the trigger, actions and business rules change.

Three Cloud Flow Types

1

Instant flow

A person deliberately selects Run, presses a button or invokes the flow from an app.

Use when the user controls the exact start time.

2

Scheduled flow

A Recurrence trigger reaches a defined time.

Use for daily, weekly or other timetable-driven work.

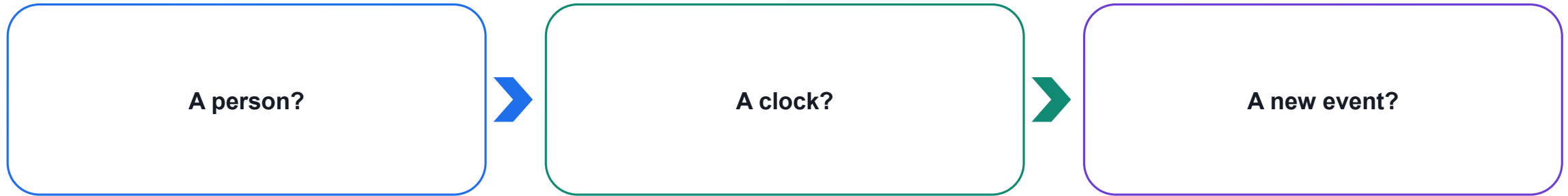
3

Automated flow

A connected-service event occurs, such as a new form response.

Use when the process must react immediately.

Choose the Flow Type by Asking One Question



Teaching point

Person = instant. Clock = scheduled. New record, form, file or message = automated.

Same Actions, Different Triggers

1

Instant

Manual trigger → send email →
add Excel row

2

Scheduled

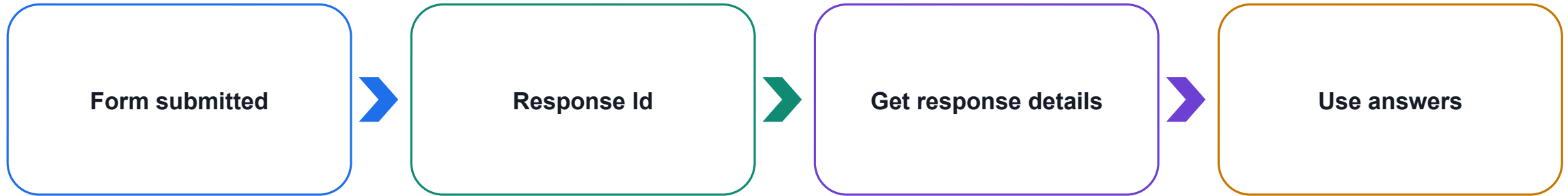
Recurrence → read pending items
→ send reminder

3

Automated

Form response → get details →
email / log / approve

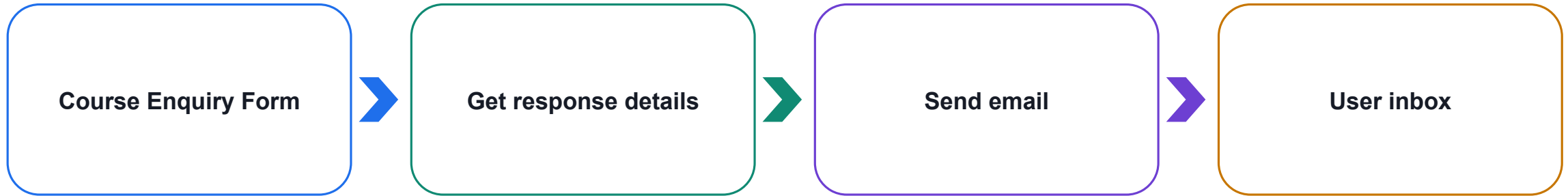
The Microsoft Forms Trigger Pattern



Teaching point

The trigger identifies the response; Get response details retrieves the actual answers.

Lab 1 — Form to Email Confirmation



Teaching point

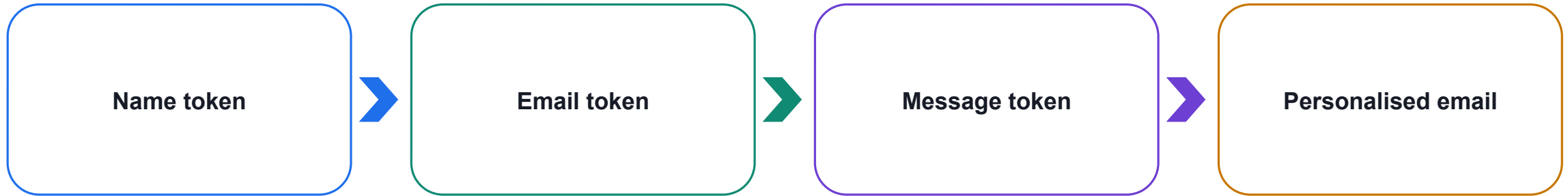
The submitted Email answer—not the maker's address—is the recipient.

Lab 1 — Design the Form

The diagram illustrates a three-step process flow, each step represented by a light blue rounded rectangle with a colored vertical bar on the left and a numbered circle in the top left corner.

- Step 1: Identity** (Blue bar and circle). The text "Name" and "Email" is listed in the bottom left.
- Step 2: Contact** (Green bar and circle). The text "Tel" is listed in the bottom left.
- Step 3: Enquiry** (Purple bar and circle). The text "Message (long answer)" is listed in the bottom left.

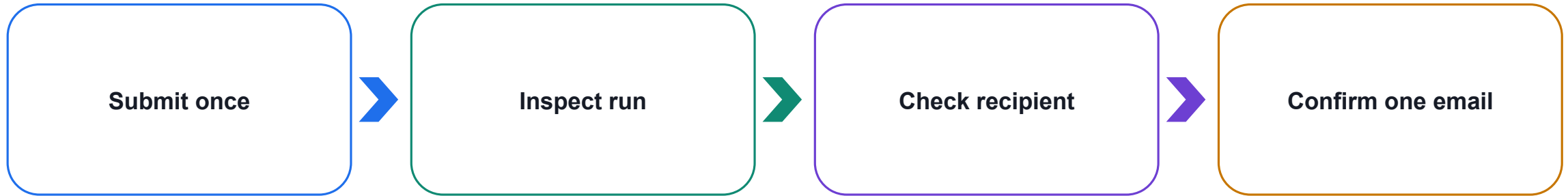
Lab 1 — Map Dynamic Content



Teaching point

Insert tokens from Get response details. Typed labels are not dynamic content.

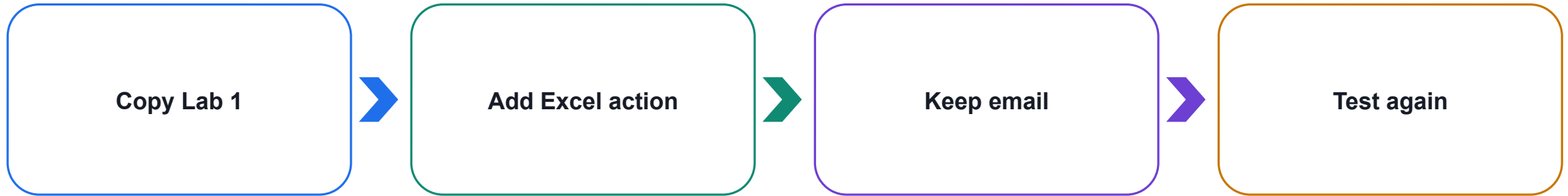
Lab 1 — Test the Real Outcome



Teaching point

One form submission should produce one successful run and exactly one message.

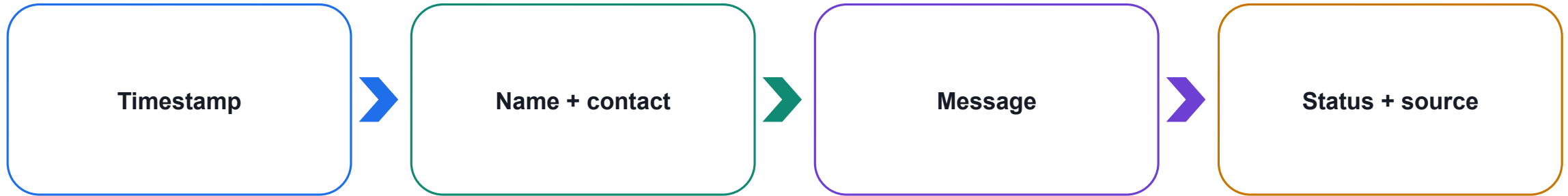
Lab 2 — Extend, Do Not Rebuild



Teaching point

Lab 2 preserves the working trigger and email, then inserts a durable audit record.

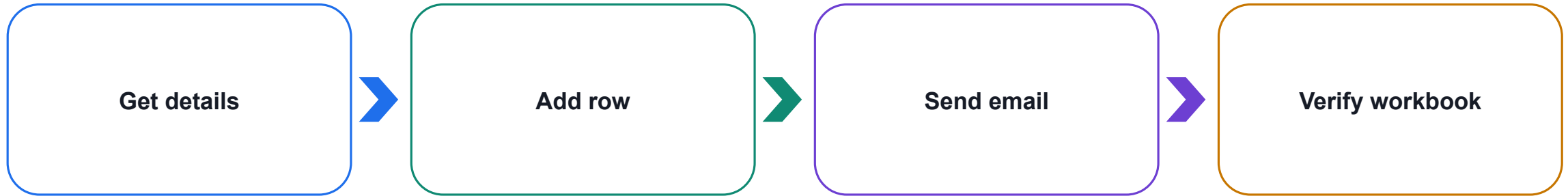
Lab 2 — Enquiry Log.xlsx



Teaching point

Excel Online actions require a named table: EnquiryLog.

Lab 2 — Action Order



Teaching point

Put logging before email when confirmation should be sent only after a successful record.

Lab 2 — Troubleshooting

1

Workbook

Store in OneDrive for Business or SharePoint.

2

Table

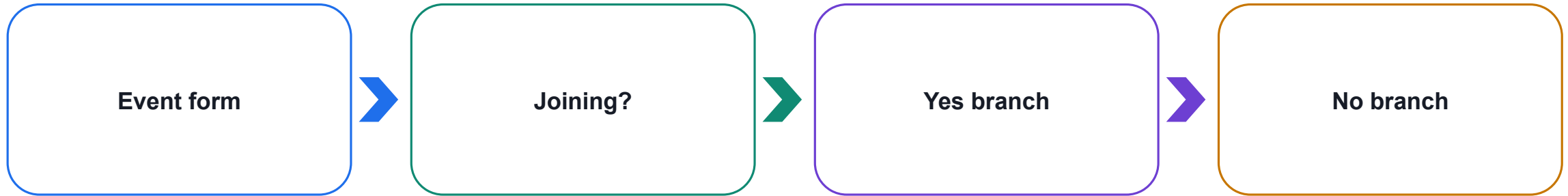
Select EnquiryLog; loose cells are not a table.

3

Lock

Close desktop Excel if the connector reports a file lock.

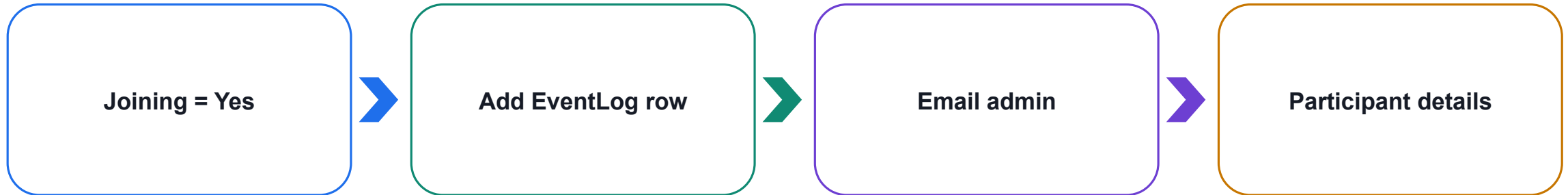
Lab 3 — Event Registration Branching



Teaching point

A choice field becomes a business rule that must be tested on both paths.

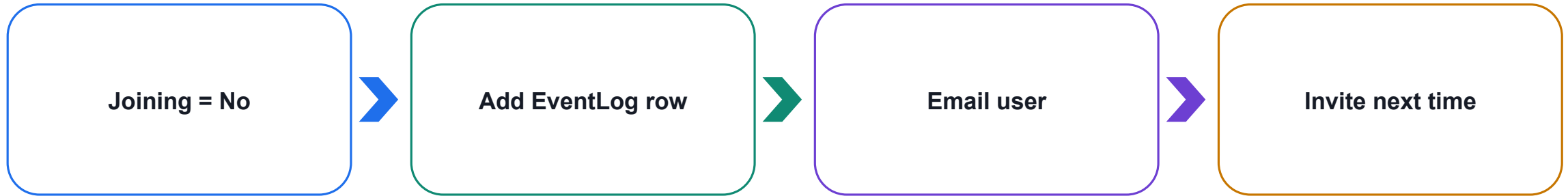
Lab 3 — Yes Branch



Teaching point

The administrator receives the person's name, email and telephone number.

Lab 3 — No Branch



Teaching point

The user receives thanks and a respectful next-event message.

Lab 3 — Branch Evidence

1

Yes test

Correct row + administrator email

2

No test

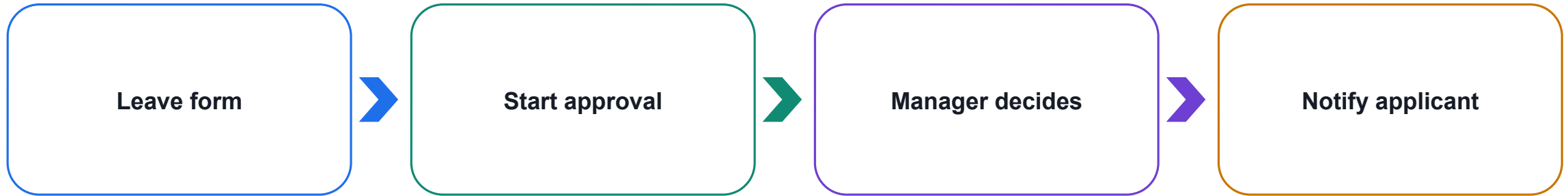
Correct row + user email

3

Audit

JoiningEvent and NotificationSent
explain the outcome

Lab 4 — Leave Application Approval



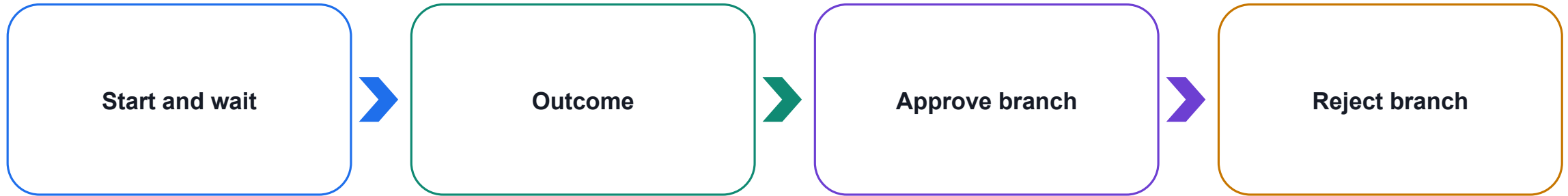
Teaching point

An automated flow can pause safely while an authorised person decides.

Lab 4 — Leave Form

1 Employee Name Email	2 Dates Leave from date Leave end date	3 Request Leave type Reason for leave
---	--	---

Lab 4 — Approval Outcome



Teaching point

Compare Outcome to Approve and include manager comments in the applicant email.

Lab 4 — Governance

1

Access

Use only authorised approvers.

2

Privacy

Limit medical and personal information.

3

Evidence

Retain decision, comments and run history.

Copilot Agent — Six Building Blocks

1

Knowledge base

Approved files, websites or SharePoint sources used to ground answers.

2

Skills

Capabilities realised through topics, tools and agent flows.

3

Tools

Connected functions that read data or create authorised outcomes.

Copilot Agent — Six Building Blocks

1

Memory

Conversation context and variables; persistence depends on enabled features and governance.

2

Model

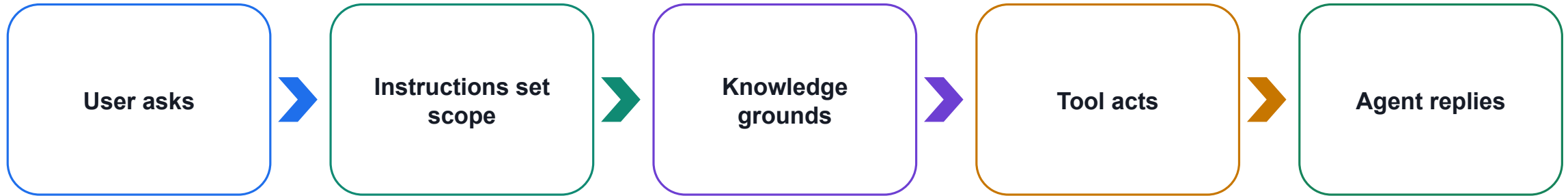
The generative model interpreting requests and composing responses.

3

System message

Instructions defining role, scope, tone, safety and escalation.

Knowledge Answers; Tools Act



Teaching point

Use static knowledge for approved explanations and tools for live, authorised actions.

Agent Safety Test Set

1

Expected

Questions the agent should answer.

2

Unsupported

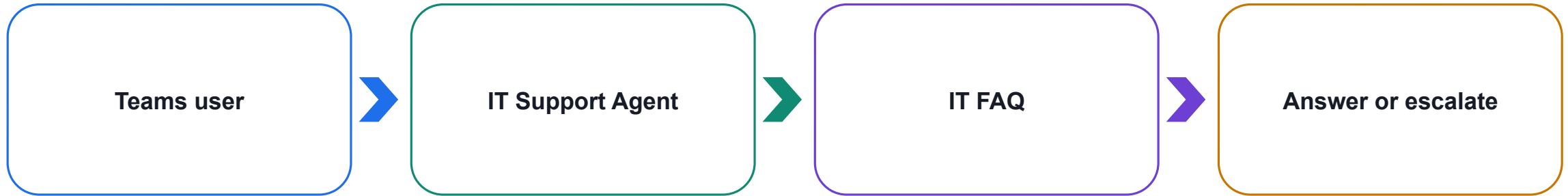
Missing or out-of-scope information.

3

Unsafe

Credentials, personal data or prohibited decisions.

Lab 5 — IT Support Agent



Teaching point

The agent never asks for passwords, MFA codes or recovery keys.

Lab 5 — Test and Deploy

1

Positive

Password reset, MFA, VPN and device loss.

2

Negative

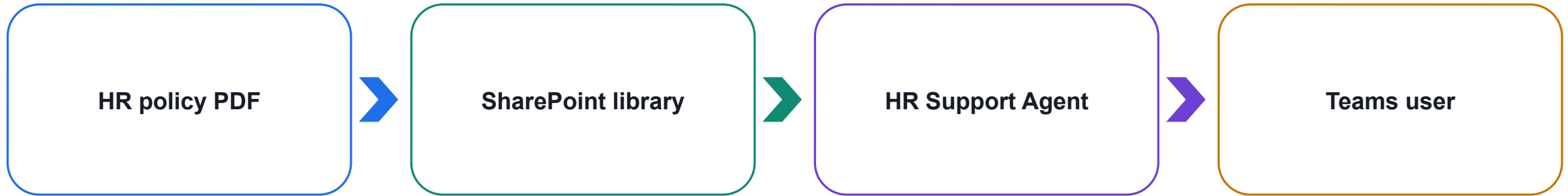
Credential request and unrelated HR question.

3

Channel

Publish latest version and verify in Teams.

Lab 6 — HR Support Agent



Teaching point

Central knowledge and source permissions support governed policy answers.

Lab 6 — HR Boundaries

1

Explain

Approved leave, expenses and workplace policy.

2

Protect

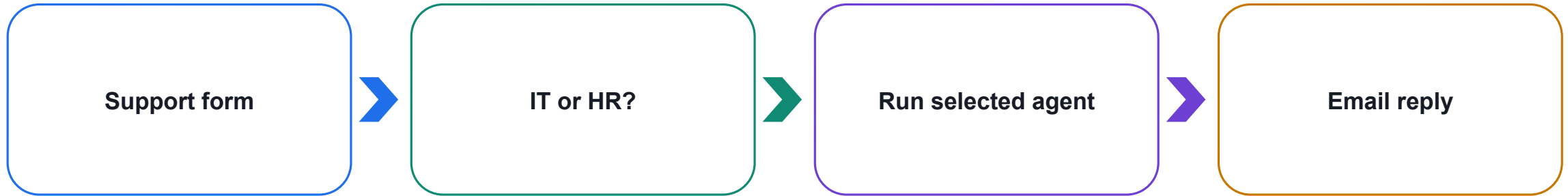
Never expose another employee's records.

3

Escalate

HR and managers make final decisions.

Lab 7 — Route Requests to Specialised Agents



Teaching point

A condition routes the same form to the specialist with the right knowledge and instructions.

Day 1 Recap

1

Flows

Forms → email → Excel →
condition → approval

2

Agents

IT FAQ and HR SharePoint
knowledge

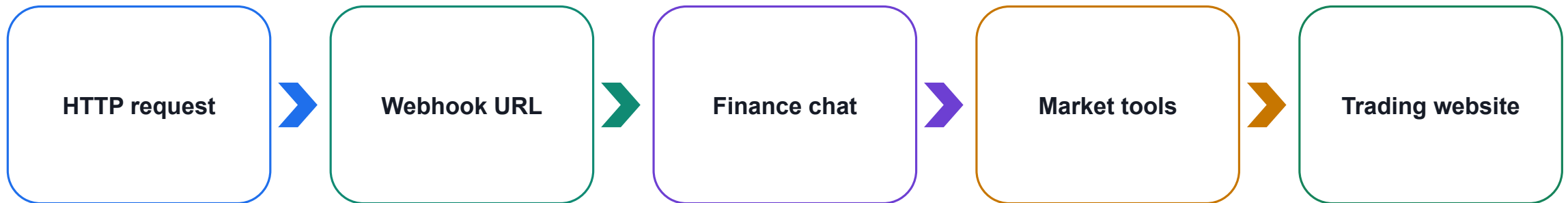
3

Integration

Form → specialised agent → user
email

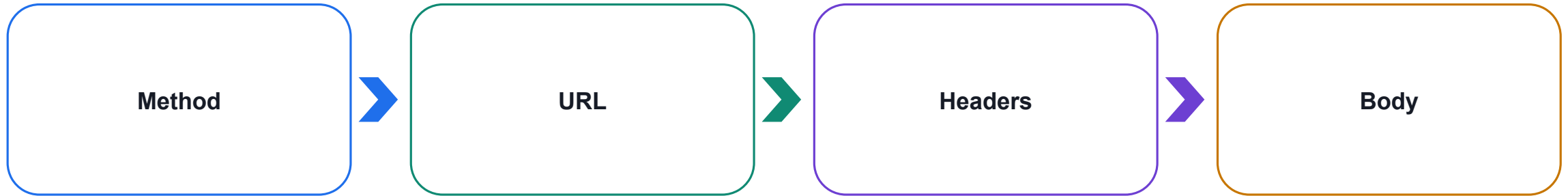
Day 2 — HTTP, Webhooks and Agent Websites

DAY 2



Expose flows safely to websites and finish with a tool-using Finance Advisor Agent.

An HTTP Request Has Four Main Parts



Teaching point

The lab websites use POST to send a JSON body to the Power Automate endpoint.

Common HTTP Methods

1

GET

Read data

2

POST

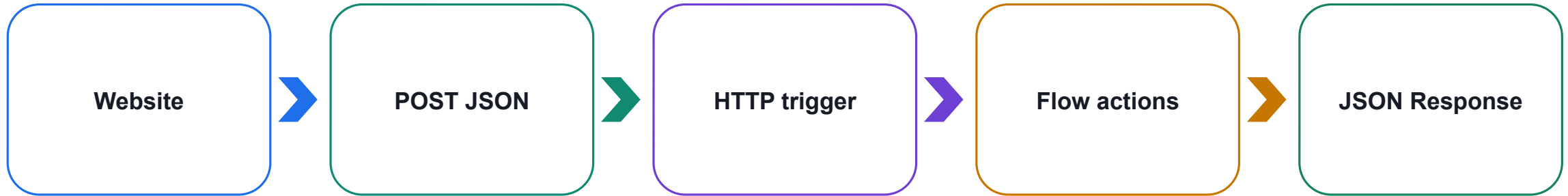
Submit data or start work

3

PUT · PATCH · DELETE

Update or remove data

A Webhook Is an Event Endpoint



Teaching point

Power Automate generates the URL after a valid trigger-and-action flow saves.

Webhook Safety

1

URL

Treat an anonymous webhook URL like a secret.

2

Input

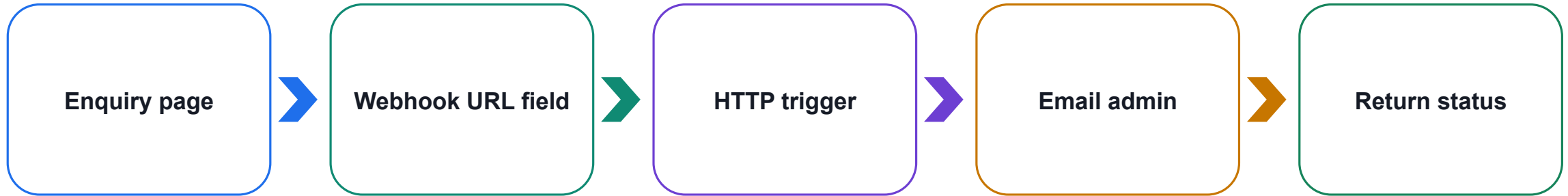
Validate, minimise and never include credentials.

3

Output

Return only the data the website needs.

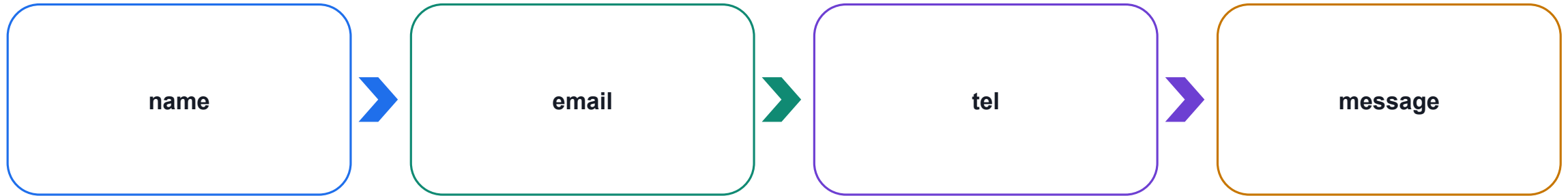
Lab 8 — Website HTTP Enquiry



Teaching point

The learner pastes the generated URL; the HTML contains no tenant endpoint.

Lab 8 — Request Contract



Teaching point

Generate the trigger schema from a sample payload before mapping fields.

Lab 8 — Complete the Flow Before Saving



Teaching point

A trigger alone is incomplete; add at least one action and resolve validation errors.

Lab 8 — Browser Test

1

Connect

Paste and save the current
webhook URL.

2

Submit

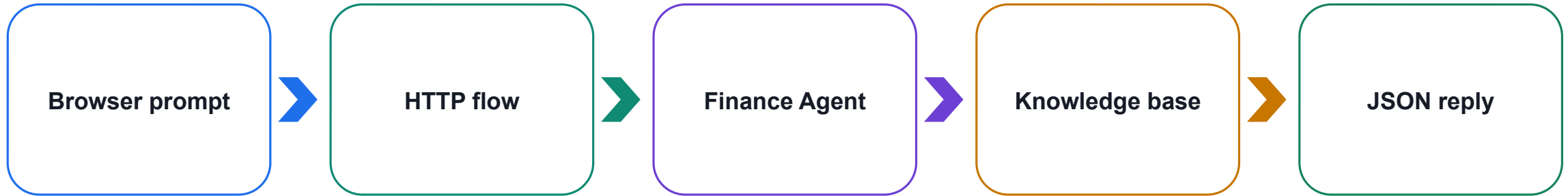
Enter realistic classroom data
once.

3

Verify

Page status, run history and
administrator email.

Lab 9 — Finance Knowledge Agent



Teaching point

The website is a channel; the agent supplies governed educational interpretation.

Lab 9 — Finance Knowledge Boundary

1

Explain

Orders, candles, timeframes,
volatility and news.

2

Qualify

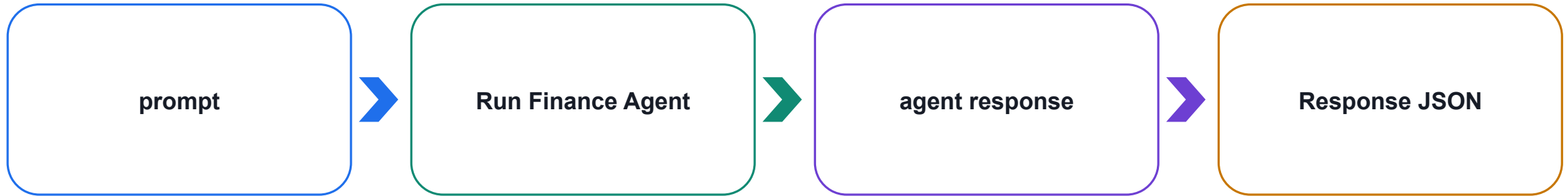
Separate facts from interpretation
and uncertainty.

3

Refuse

No guarantees or personalised
buy/sell instructions.

Lab 9 — HTTP-to-Agent Flow



Teaching point

Insert the actual agent-response output token into the JSON reply.

Lab 9 — Test Set

1

Concept

Market versus limit order.

2

Uncertainty

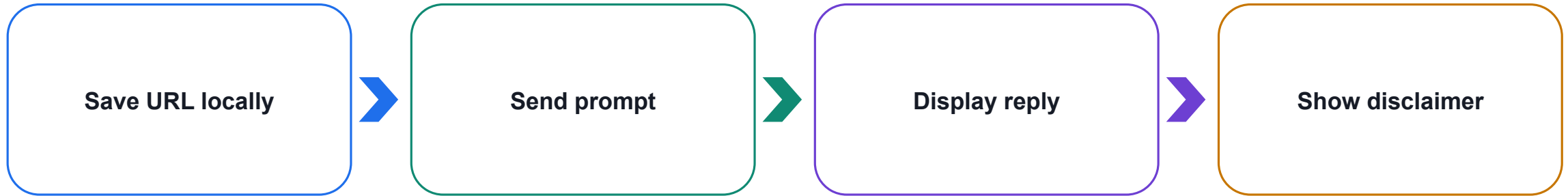
Question not covered by knowledge.

3

Boundary

Request for guaranteed returns.

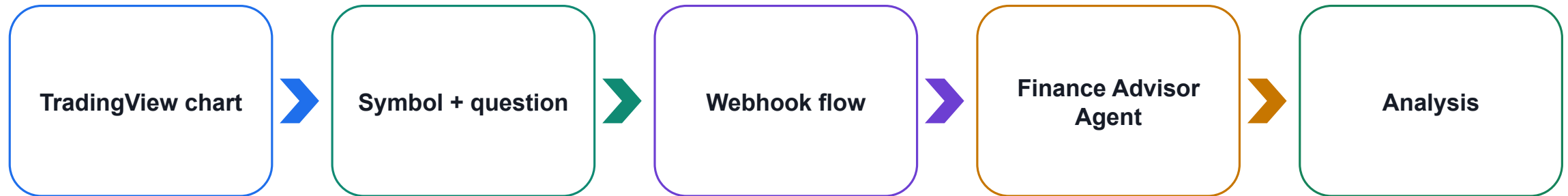
Lab 9 — Connect the Chat Page



Teaching point

One message should create one flow run; do not embed tenant URLs in source.

Lab 10 — Trading Advisor Capstone



Teaching point

The experience combines a market chart with governed multi-timeframe and news analysis.

Lab 10 — Four Authorised Tools

1

Twelve Data

1-minute candles
15-minute candles
1-hour candles

2

NewsAPI

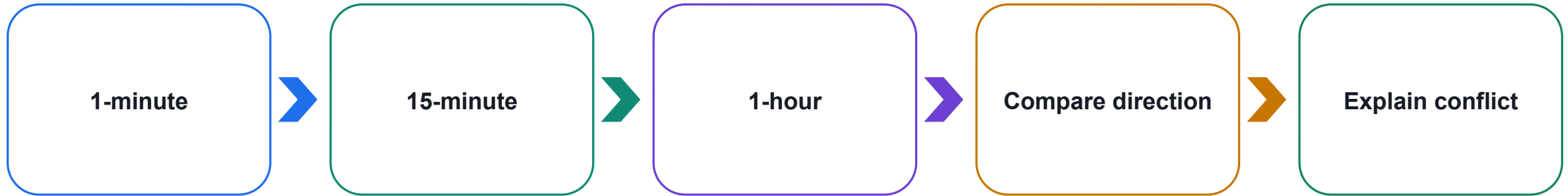
Recent relevant company or
symbol news

3

Guardrails

Validate symbols, limits, errors and
credentials

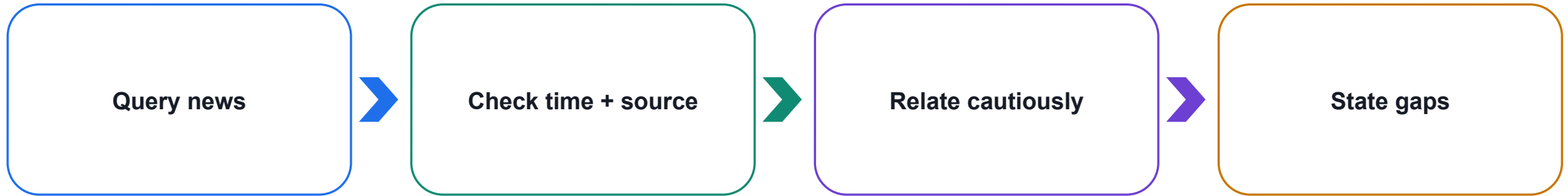
Lab 10 — Multi-Timeframe Reasoning



Teaching point

Short intervals contain more noise; a sound response preserves conflicting evidence.

Lab 10 — Add News Context



Teaching point

A headline can be delayed, incomplete or already reflected in price.

Lab 10 — Secure API Design

1

Store

Protected connections,
environment variables or custom
connector.

2

Limit

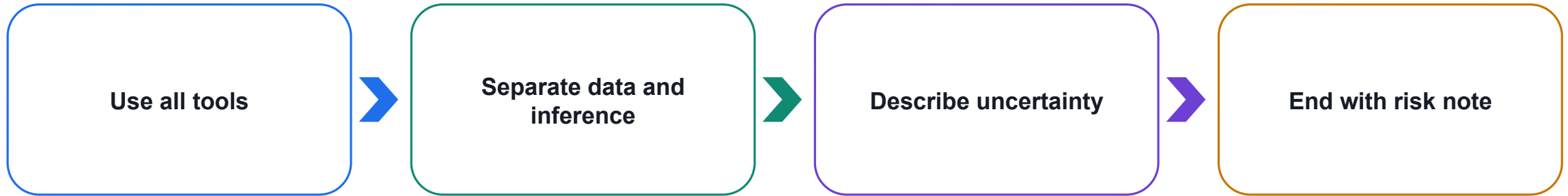
Small result sets and provider rate
limits.

3

Hide

Never expose keys in HTML,
prompts, logs or screenshots.

Lab 10 — Finance Advisor Instructions



Teaching point

The agent provides education; it never promises returns or tells a person what to buy.

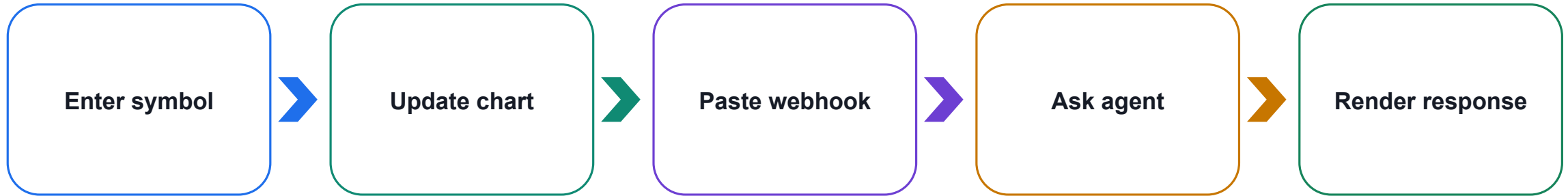
Lab 10 — HTTP Request Contract



Teaching point

Reject invalid symbols and return a clear error instead of fabricated analysis.

Lab 10 — TradingView Website



Teaching point

TradingView visualisation and agent analysis are related but independent components.

Lab 10 — Test Matrix

1

Valid

All data tools succeed and output is qualified.

2

Partial

Missing news or conflicting timeframes are disclosed.

3

Invalid

Bad symbol or advice request gets a safe response.

End-to-End Agentic Architecture



Teaching point

Each boundary has a distinct responsibility and an auditable input/output.

Operational Readiness Review

1

Security

Authentication, secrets and least privilege.

2

Reliability

Timeouts, retries, rate limits and fallbacks.

3

Governance

Approved sources, disclaimers, logs and owners.

Day 2 Recap

1

Connect

HTTP request + learner-entered
webhook URL

2

Ground

Finance knowledge + live
authorised tools

3

Respond

Structured JSON + uncertainty +
risk reminder

What You Can Automate Next

Onboarding: new-hire checklists and welcome emails

Reporting: scheduled data pulls and summaries

Helpdesk: triage and route incoming requests

Document handling: filing, approvals and reminders

- Anywhere people copy data or chase approvals by hand

Common Pitfalls to Avoid

Automating a broken process — fix the process first

Trying to build everything in one giant flow

Skipping tests, then surprises in production

No error handling — failures go unnoticed

- Vague agent prompts that return inconsistent data

Common Errors & Quick Fixes

“Unauthorized” sending email — reconnect Outlook with a mailbox-enabled account

Approval ‘valid users’ error — pick the approver from the dropdown (a real user in your tenant)

Date shows literal utcNow() — enter it via the fx / Expression editor so it becomes a token

Excel cell shows ##### — the column is just too narrow; auto-fit it

- Unwanted ‘For each’, or agent can’t see its flow — use single-value fields; same environment in both tools

Governance & Security Basics

Use connections and permissions you're authorised to use

Keep sensitive data out of emails and logs where possible

Follow your organisation's data and approval policies

Document what each flow does and who owns it

- Review and retire flows that are no longer needed

Rolling This Out at Work

Pick one painful, repetitive task to automate first

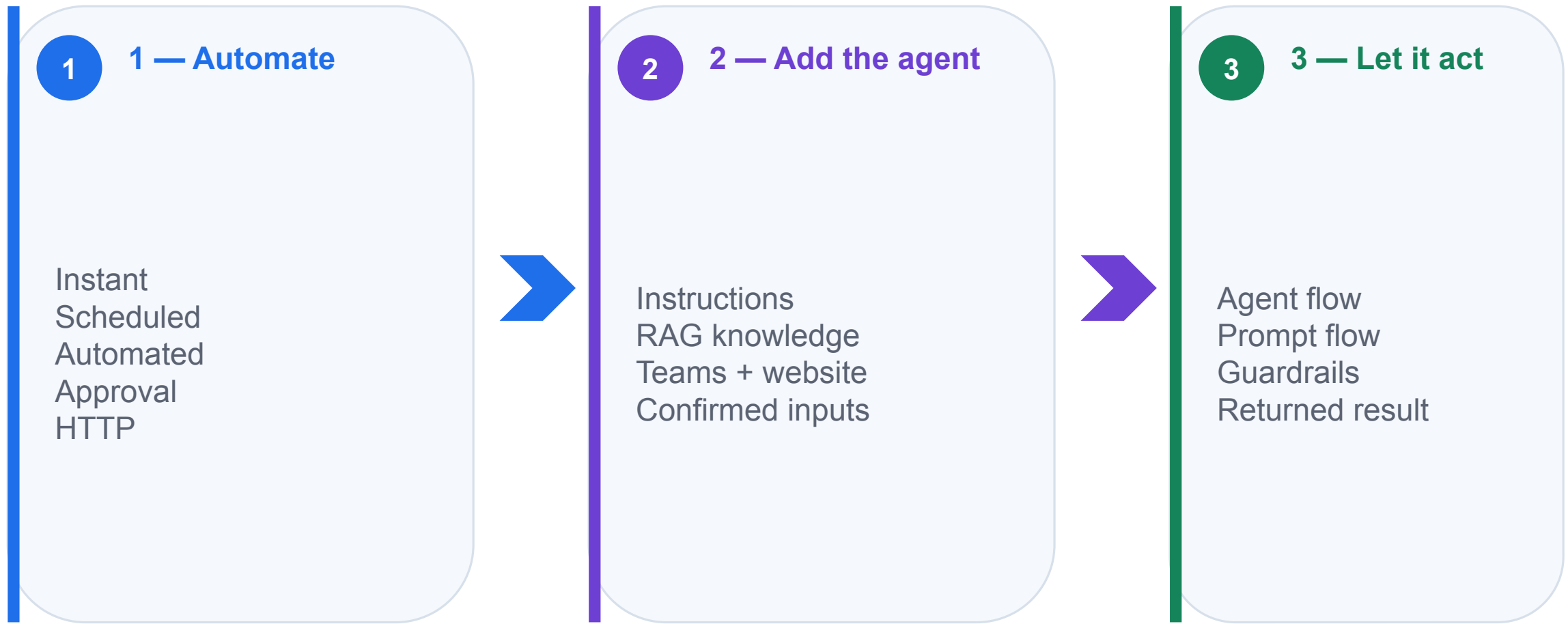
Prove the value, then share the win with your team

Build a small library of reusable flows and agents

Involve the people who do the work in the design

- Iterate — improve flows as needs change

Course Recap — The Big Picture



Key Takeaways

Every workflow is Trigger → Actions → Output

Power Automate does the work; Copilot Studio adds the intelligence

Structured agent outputs make flows reliable

Orchestration turns steps into complete processes

- Start small, test, and improve — then scale

Resources & Where to Go Next

make.powerautomate.com — build and manage your flows

copilotstudio.microsoft.com — create and publish agents

Microsoft Learn — free guided learning paths

Templates gallery — start from pre-built flows

- Your community — share flows and patterns with colleagues

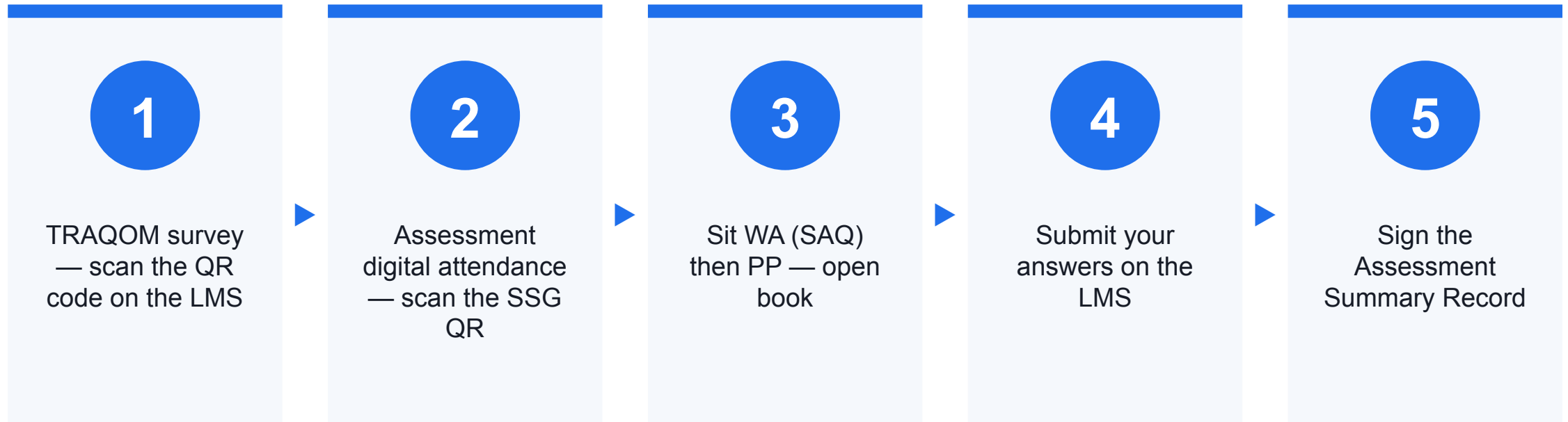
Courseware & Assessment on the LMS

- Download the slides and the Learner Guide from the LMS for the open-book assessment.
- Portal: <https://lms-tms.tertiaryinfotech.com/>
- Submit your Written Assessment and Practical Performance answers on the LMS.
- Complete the TRAQOM survey via the QR code on the LMS.

Assessment

- Written Assessment (SAQ) — 1 hour. Practical Performance (PP) — 1 hour.
- Open book: slides, Learner Guide and approved materials only.
- Remember to take the Assessment digital attendance (TRAQOM).
- Submit your completed answers on the LMS at <https://lms-tms.tertiaryinfotech.com/>.

Assessment Flow



Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Thank You

Now go automate something — questions welcome